Algorithm: Configurable Payment Processing System Using Strategy Pattern

1. Start the program.

2. Define a PaymentStrategy interface with a pay(amount) method.

3. Create three payment classes: CashPayment, CardPayment, and UPIPayment.

4. Each payment class implements the pay(amount) method in its own way.

5. Create a PaymentProcessor class.

6. Initialize the payment strategy as None.

7. Read the payment amount from the user.

8. Display the payment options: 1→ Cash, 2→ Card, 3→ UPI.

9. Read the user's choice.

10. If choice = 1, set the strategy to CashPayment.

11. Else if choice = 2, set the strategy to CardPayment.

12. Else if choice = 3, set the strategy to UPIPayment.

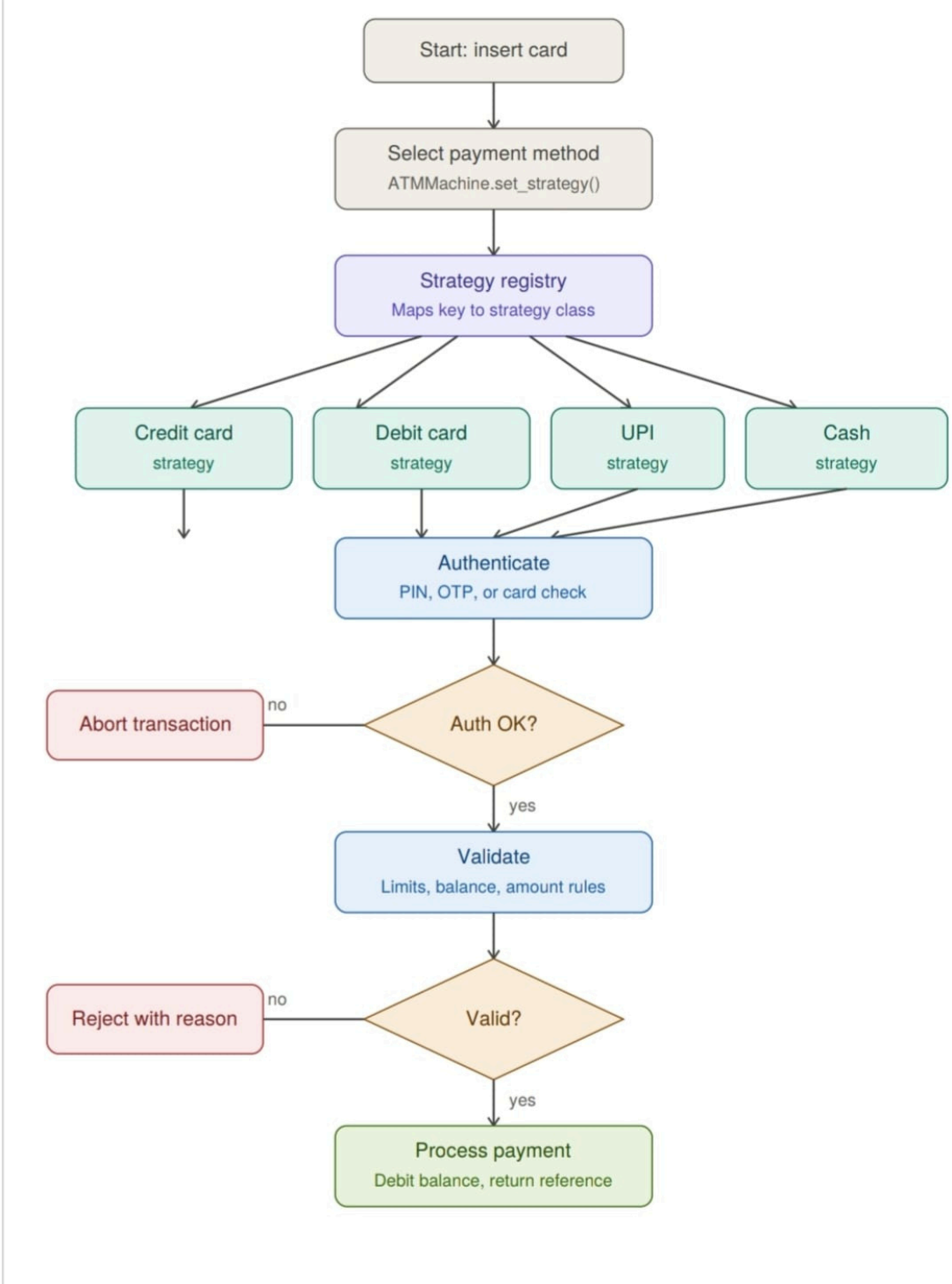
13. Otherwise, display 'Invalid Choice' and terminate the program.

14. Call the process_payment(amount) method.

15. The selected payment strategy executes the pay(amount) method.

16. Display the payment success message.

17. End the program.



Strategy Pattern Implementation - Python Source Code

```
from abc import ABC, abstractmethod


# Strategy Interface

class PaymentStrategy(ABC):
```

```
    @abstractmethod
    def pay(self, amount):
        pass


# Concrete Strategies
class CreditCardPayment(PaymentStrategy):
    def pay(self, amount):
        print(f"Paid ₹{amount} using Credit Card")


class DebitCardPayment(PaymentStrategy):
    def pay(self, amount):
        print(f"Paid ₹{amount} using Debit Card")


class UPIPayment(PaymentStrategy):
    def pay(self, amount):
        print(f"Paid ₹{amount} using UPI")


# Context Class
class PaymentProcessor:
    def __init__(self, holder_name, account_number, password):
        self.holder_name = holder_name
        self.account_number = account_number
```

```
        self.password = password
        self.strategy = None

    def set_payment_strategy(self, strategy):
        self.strategy = strategy

    def process_payment(self, amount, entered_password):
        if entered_password != self.password:
            print("Invalid password! Payment failed.")
            return

        print("\nAccount Holder:", self.holder_name)
        print("Account Number:", self.account_number)

        if self.strategy:
            self.strategy.pay(amount)
        else:
            print("No payment method selected!")


# Main Program
holder_name = input("Enter account holder name: ")
account_number = input("Enter account number: ")
password = input("Set account password: ")

processor = PaymentProcessor(holder_name, account_number, password)
```

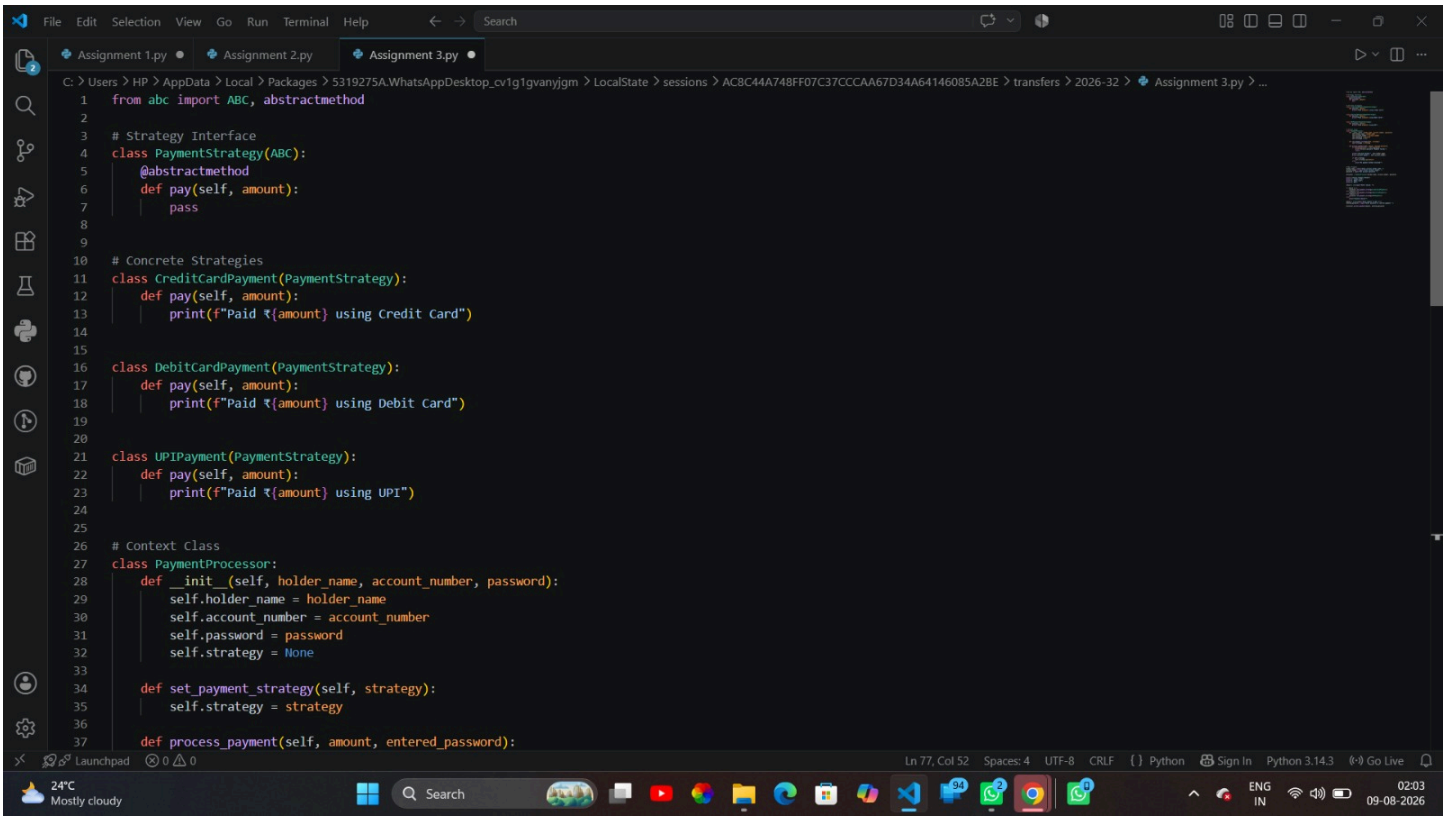
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print("\nSelect Payment Method")
print("1. Credit Card")
print("2. Debit Card")
print("3. UPI")

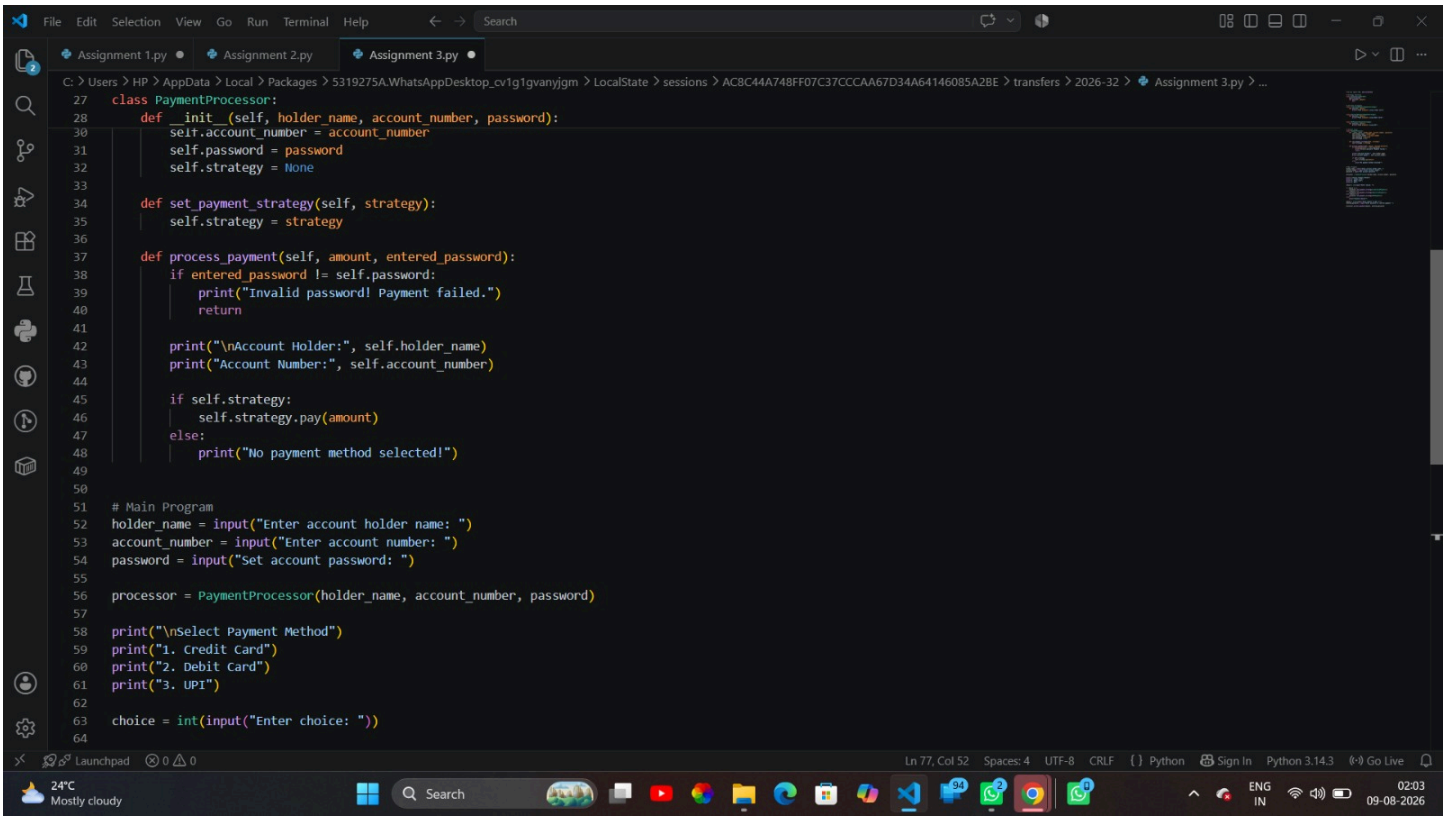
choice = int(input("Enter choice: "))

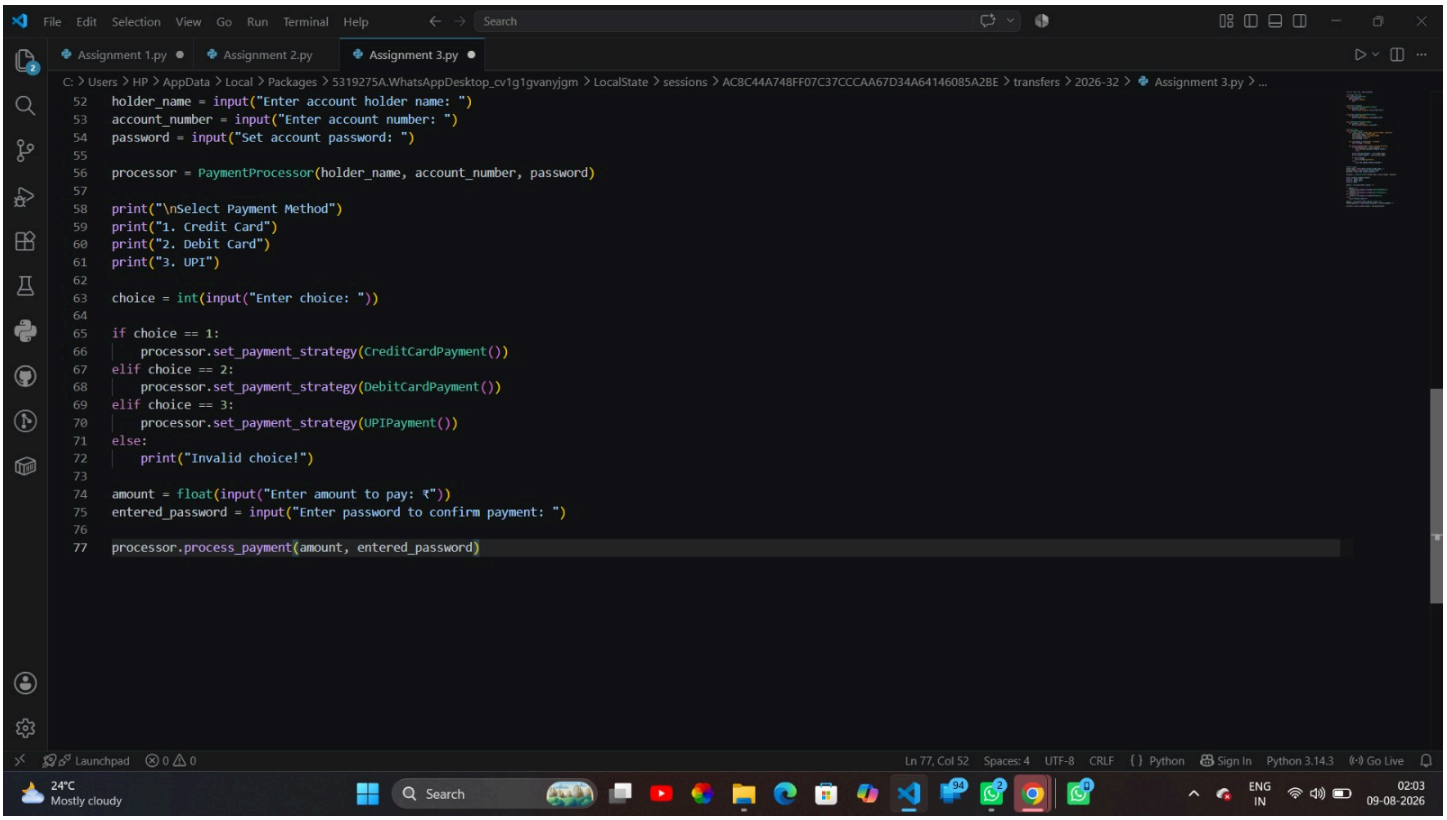
if choice == 1:
    processor.set_payment_strategy(CreditCardPayment())
elif choice == 2:
    processor.set_payment_strategy(DebitCardPayment())
elif choice == 3:
    processor.set_payment_strategy(UPIPayment())
else:
    print("Invalid choice!")

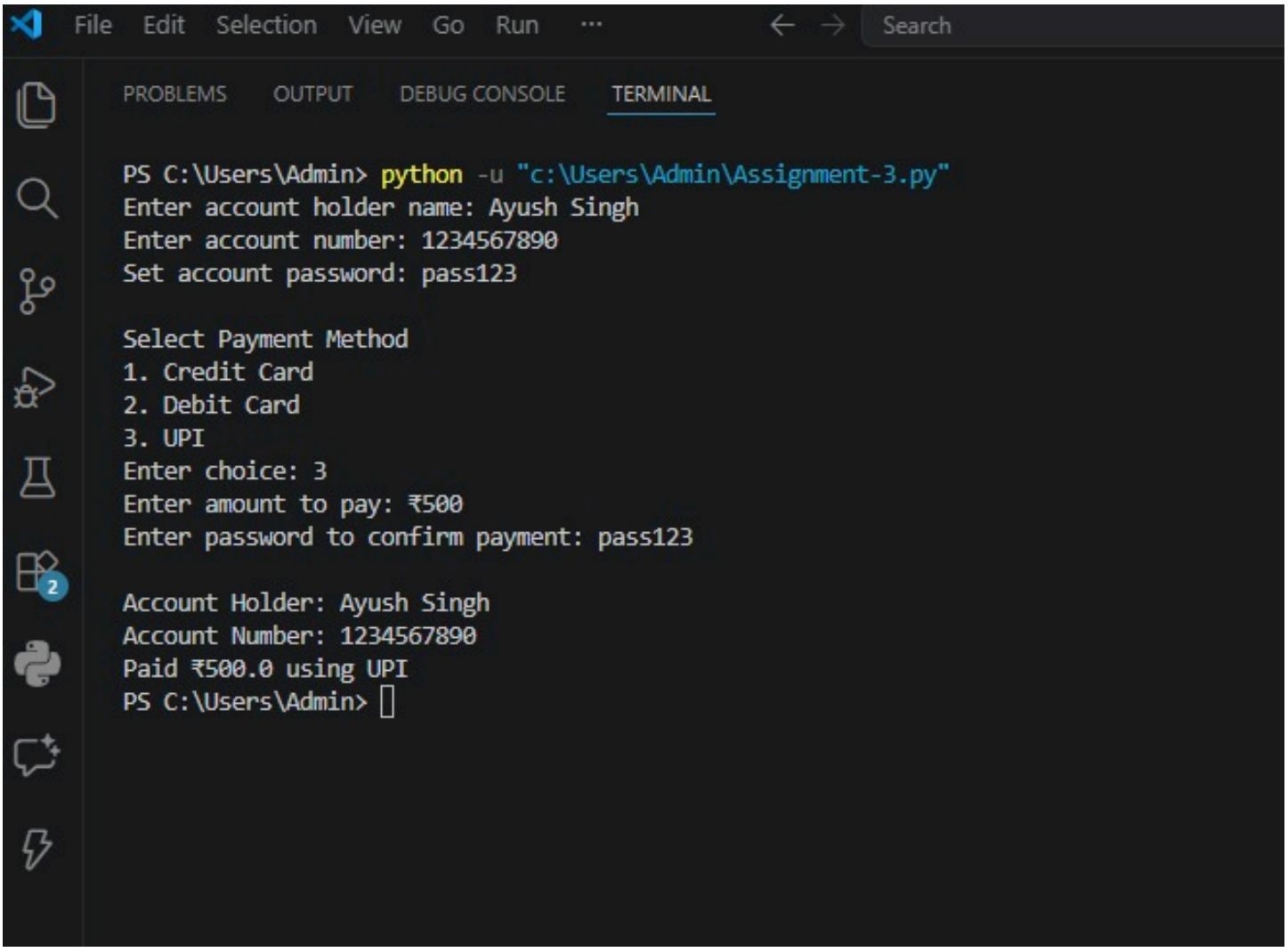
amount = float(input("Enter amount to pay: ₹"))
entered_password = input("Enter password to confirm payment: ")

processor.process_payment(amount, entered_password)
```


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Assignment 3.py
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from abc import ABC, abstractmethod
Strategy Interface
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Concrete Strategies
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self.strategy = strategy
def process_payment(self, amount, entered_password):
if entered_password != self.password:
print("Invalid password! Payment failed.")
return
print("\nAccount Holder:", self.holder_name)
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if self.strategy:
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Main Program
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PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PS C:\Users\Admin> python -u "c:\Users\Admin\Assignment-3.py"
Enter account holder name: Ayush Singh
Enter account number: 1234567890
Set account password: pass123
Select Payment Method
1. Credit Card
2. Debit Card
3. UPI
Enter choice: 3
Enter amount to pay: ₹500
Enter password to confirm payment: pass123
Account Holder: Ayush Singh
Account Number: 1234567890
Paid ₹500.0 using UPI
PS C:\Users\Admin>